Lecture 4: Option Pricing

4.1 Black-Scholes

Under the risk-neutral measure, the call price is **$C = S\,N(d1) - K\,e^{-rT}\,N(d2)$**, where $d1 = [\ln(S/K) + (r + \sigma^2/2)T] / (\sigma\sqrt{T})$.

- Assumption: constant volatility σ
- Assumption: no dividends

4.2 The Greeks

Greek	Formula	Interpretation
Delta	$N(d1)$	Hedge ratio
Gamma	$\varphi(d1)/(S\sigma\sqrt{T})$	Convexity
Vega	$S\,\varphi(d1)\,\sqrt{T}$	Vol sensitivity

4.3 Exercise

1. Compute Δ for $S=100$, $K=105$, $\sigma=20\%$, $r=3\%$, $T=0.5$.
2. Explain why Γ peaks ATM.